MILITARY SPECIFICATION

TEST SET, COMPUTER, TS-909( )/PPM

1. SCOPE

1.1 Scope.- This specification covers one type of null detecting device used to aline and adjust 400-cycle analog computers, designated as Test Set, Computer TS-909( )/PPM. (See 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
PPP-B-636	Boxes, Fiber
PPP-F-320	Fiberboard, Sheet, Stock and Cut Shapes
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-V-173	Varnish, Moisture and Fungus-Resistant, For the Treatment of Communications, Electronic and Associated Electrical Equipment
MIL-W-6858	Welding, Aluminum, Magnesium, Non-hardening Steels, or Alloys and Titanium, Spot, and Stitch
MIL-M-10578	Metal Conditioner and Rust Remover
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-901	Shockproof Equipment, Class HI

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STANDARDS

MIL-STD-22	Welded-Joint Designs.
MIL-STD-105	Sampling Procedures and Tables for Inspection of Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme Temperature Cycle.
MIL-STD-170	Moisture-Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical. Defects.

DRAWINGS

SIGNAL CORPS

SC-DL-199738	Test Set, Computer TS-909( )/PPM.
SC-DL-199746	Cable Assembly, Power Electrical CX-3382( )/U.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Construction. - Construction of the equipment on contract shall be in accordance with the drawings and documents listed in Signal Corps Drawing and Data List SC-DL-199746 and SC-DL-199738.

3.2 Preproduction samples. - The contractor shall furnish preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3.)

3.3 Parts, materials, and processes; general. -

3.3.1 Cleaning. -

3.3.1.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

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3.3.1.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.3.2 Soldering. -

3.3.2.1 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.3.2.2 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.3 Welding. - Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858.

3.3.3.1 Cleaning prior to welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.3.3.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium or stainless steel.

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3.3.3.3 Cleaning after welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.3.4 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.3.5 Connections; general. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.3.6 Marking. -

3.3.6.1 General. - Marking shall conform to Specification MIL-M-13231. (See 4.4.) Front panel marking and the equipment drawings shall be group I as described in that specification.

3.3.6.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.3.6.3 Serial number. - Each Test Set, Computer TS-909( )/PPM shall be serial numbered.

3.3.7 Finish. - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4.)

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3.4 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand picked for fit or performance; however, matched sets, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc. (See 4.10.)

3.5 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7.

3.6 Service conditions. - The equipment shall meet the following service conditions --where a test is referenced, meeting the test shall be considered as compliance with the requirements:

3.6.1 Operation. - Continuous use for a period of one year, under the worldwide environmental conditions specified in 3.6.2 through 3.6.7, with a duty cycle of six hours on and two hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.6.2 Temperature. (See 4.6.1.) -

(a) Operating: Ambient temperature in the range of +150° F to -40° F. (The 150° F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160° F to -80° F; exposure at the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.6.3 Relative humidity. (See 4.6.2.) - Daily exposure up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for four hours.

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3.6.4 Elevation. (see 4.6.3). -

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.6.5 Rain. (See 4.6.5). - Two inches of rainfall per hour, with wind at 20 miles per hour.

3.6.6 Vibration and bounce. -

3.6.6.1 Vibration, internal. - The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.6.6.)

3.6.6.2 Shock, ballistic. - The unit shall be operable and shall meet the tests of 4.8.2 after subjection to the shock test of 4.6.4. Any physical damage shall be minor only.

3.6.6.3 Immersion. (See 4.6.7). - Three feet of water for 2 hours. (With cover in place.)

3.7 Electrical requirements. -

3.7.1 Power source. - The equipment shall operate from a power source delivering the following:

(a) 115 volts ± 10%, 400-cps ± 5% single phase, at five amperes minimum.

(b) 220 volts ± 10% at 10 milliamperes (ma) direct current minimum. (See 4.8.1.)

3.7.2 Sensitivity and balance. -

3.7.2.1 Balance. - When tested as specified in 4.8.2(d) adjustment of the balance adjustment potentiometer shall cause the NULL METER to deflect to the right and left of zero at least 25 percent of full scale.

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3.7.2.2 Sensitivity.- Unbalancing of the balanced test set by the application of 400-cps input signals shall cause deflection of the NULL METER as indicated below: (See 4.8.4.)

Input voltage	Meter deflection
0.2 millivolts	Noticeable deflection
5 millivolts	Half-scale deflection

3.7.2.3 Voltage divider.- The Kelvin Varley type voltage divider shall have the following characteristics: (See 4.8.2.)

(a) Max resistance - 10,000 ohms ± 05%.

(b) Min. resistance - no higher than 0.2 ohms (see 4.8.2.)

3.8 Final operation.- Just prior to packaging, the equipment shall be examined visually and electrically. The electrical examination shall be performed without any signal input. (See 4.9.)

3.9 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.1.)

3.10 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portion of the following paragraphs herein:

3.1	Construction	3.3.3	Welding
3.3.1	Cleaning	3.3.6	Marking
3.3.2	Soldering	3.3.7	Finishing

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. - The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5.)

(2) Acceptance inspection of preparation for delivery. (See 4.12.)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B. (See tables II and III, respectively.) The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I. - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3.)	Req para	Insp para
Temperature	3.6.2	4.6.1
Moisture-resistance	3.6.3	4.6.2
Elevation	3.6.4	4.6.3
Rain	3.6.5	4.6.5
Vibration	3.6.6.1	4.6.6
Shock	3.6.6.2	4.6.4
Immersion	3.6.6.3	4.6.7

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4.4 Acceptance inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Marking	3.3.6
Finish	3.3.7

4.5 Acceptance inspection of equipment before preparation for delivery. - The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4, 4.5.1 and 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, and group B inspection shall be preconditioned after final assembly. (See 3.5.)

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table II. - Group A inspection

Inspection	Req para	Insp para	AQL	
			Major	Minor
Visual and mechanical - - -	3.10	4.11	1.5 dphu	6.5 dphu **
Electrical ⊛				
Power source - - - -	3.7.1	4.8.1)	2.5 dphu combined for the group	
Balance - - - - - -	3.7.2.1	4.8.3)		
Sensitivity - - - - -	3.7.2.2	4.8.4)		
Voltage divider - - -	3.7.2.3	4.8.2)		
Final operation - - -	3.8	4.9	1%	

⊛ All electrical defects are considered to be major.
⊛⊛ dphu = defects per hundred units.

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4.5.2 Group B inspection. - This inspection, including sampling, shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The group B sampling plan shall be L-7 for normal and tightened inspection, and L-5 for reduced inspection.

Table III. - Group B inspection

Inspection	Req para	Insp para	AQL
Interchangeability - - -	3.4	4.10	6.5. %

4.6 Service conditions tests. -

4.6.1 Temperature test. - The equipment shall be subjected to the temperature cycle shown in Standard MIL-STD-169. Performance shall be as follows:

Steps	Paragraph	Measurement	Performance
1, 3 and 7	4.6.2.3	Operation	Full specification requirements

4.6.2 Moisture-resistance test. -

4.6.2.1 Test conditions. -

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start measurements not more than five minutes after power is applied. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

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4.6.2.2 Test method. - The equipment shall be tested as follows:

(a) Dry at 130° ± 5° F for 24 hours.

(b) Condition at 77° ± 5° F and 40 to 50 percent relative humidity for 24 hours.

(c) Make measurements as designated in 4.6.2.3, and readjust or realine if necessary to meet specification requirements. Performance shall be full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.6.2.3 shall be made during the periods shown on the standard and shall comply with full specification requirements.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specification requirements for those measurements specified in 4.6.2.3.

4.6.2.3 Operation. - Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.8.1	Power source	Full specification requirements
4.8.2	Balance and Sensitivity	

4.6.2.4 Failure. - If the equipment fails to meet the performance specified in 4.6.2.3 during cycling, it will be realined or readjusted once. If the equipment then fails to meet 4.6.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.6.2.2 (e), it does not pass the test.

4.6.3 Elevation test. - The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

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(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurement:

Paragraph	Measurement
4.6.2.3	Operation

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurement specified in 4.6.3 (a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for one hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurement specified in 4.6.3 (a). The equipment shall meet specified performance for this measurement.

4.6.4 Shock test; ballistic.- The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The equipment shall be mounted in such a way as to approximate the actual rigidity encountered in service. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

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4.6.5 Rain test. - The test set minus the cover shall be tested as follows:

(a) Dry at 150° ± 5°F for 48 hours.

(b) Condition for four hours at 77° ± 5°F and 40 to 60 percent relative humidity.

(c) Take the measurements specified in 4.8.2 to insure that the equipment meets the specified performance.

(d) Place the equipment, in its normal operating position, in a test chamber maintained at 77° ± 5°F. Allow simulated rain to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ± 5°F and shall fall freely at a rate of two inches per hour. Maintain this condition for one hour.

(e) With the rainfall continuing at the two-inch rate introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for one hour.

(f) Rotate the equipment 90 degrees hotizontally to expose a second side of the component to the wind and the rain. Maintain this condition for one hour.

(g) Continue as in (f) until all four sides of the equipment have been exposed to the wind and rain each side for one hour.

(h) Immediately after completion of the foregoing repeat the measurements specified in (c) without removing the equipment from the chamber but with the wind and the rain discontinued. Complete the measurements within one hour. The equipment shall meet the specified performance.

(i) Open the equipment. There shall be no visible evidence of leakage.

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4.6.6 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.6.6.1.

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.6.7 Immersion test.- The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ± 2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and the cases wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.7 Bounce preconditioning.- The unit, with shock mounts removed or blocked, shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of three inches as the unit bounces. The unit shall be subjected to the preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.8 Electrical tests.-

4.8.1 Power source.- The equipment shall be operated with a power source having a range of supply line voltages from 103.5 to 126.5 and a range of frequencies from 380 to 420 cps t to determine conformance to 3.7.1.

4.8.2 Voltage divider test.- The voltage divider shall be measured for conformance to the resistance requirements of 3.7.2.3.

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4.8.3 Balance.- The equipment shall be tested as follows to determine conformance to 3.7.2.1:

(a) Apply 115 volts, 400-cps to terminals H and E of input jack 4601 on the test set. Apply + 220 volts dc to terminal A of the input jack and dc ground to terminal C.

(b) Switch on the Computer Test Set.

(c) Set the BRIDGE ADJ dials to zero.

(d) Short the TEST PROBE, + REF VOLT, and -REF VOLT terminal to the GROUND terminal. Rotate the balance adjustment potentiometer (R4608) first fully clockwise and then fully counterclockwise. Observe the NULL METER needle deflection. Return the NULL METER to zero. (See 3.7.2.1.)

(e) Return the NULL METER needle to zero and remove the shorts.

4.8.4 Sensitivity.- The equipment shall be tested as follows to determine compliance with 3.7.2.2:

(a) Connect the Test Probe, the -Ref. volt lead and the ground lead to a common terminal which is one side of a 6.3 volt, 400 cycle input signal. Connect the +Ref. volt lead to the other terminal side of the input signal.

b. Set the Bridge Adjust Dials to zero and zero the Null Meter Needle with the balance pot. (R4608) if necessary.

c. Advance the inner dial to 0.1 (the first engraved line past zero). This should cause a noticeable deflection of the Null Meter Needle. Reverse the input signal leads: the null meter needle shall deflect the same amount in the opposite direction.

d. Advance the inner dial to 1.0, this should cause at least 1/2 scale deflection of the Null Meter Needle.

e. Return the input signal amplitude to 1 volt and the Bridge Adjust dials to zero. Measure the voltage across terminals 1 and 2 of the precision potentiometer. Advance the Bridge Adjust Dial for minimum noticeable Null Meter deflection--the change in voltage across terminals 1 and 2 should not exceed 0.2 mv. Advance the Bridge Adjust Dial for 1/2 scale Null Meter deflection--the change in voltage should not exceed 5 mv. Adjust the Bridge Adjust dial for full scale deflection--the change in voltage should not exceed 22 mv.

f. Reconnect the leads and probe as follows: The Test Probe to one input signal supply terminal and the -Ref. volt lead to the other input signal supply terminal. The ground lead should be connected to the + Ref. volt lead.

g. Apply a 5 mv. input signal to the supply terminals. This signal should cause a noticeable Null Meter Needle deflection for a signal of either polarity.

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4.9 Final operational test. - Prior to packaging, the equipment shall be checked visually for damaged parts, loose parts and completeness. All electrical circuits shall be checked for apparent malfunction.

4.10 Inspection for interchangeability. - The equipment shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.4. Specific points that shall be gaged or measured are as follows:

(a) External dimensions of equipment case.

(b) Internal dimensions of equipment cover.

(c) Location of hinges and fasteners on cover and case.

4.11 Visual and mechanical inspection. - The units shall be examined for defects listed in Standard MIL-STD-252 and table IV.

Table IV. - Classification of visual and mechanical defects

Classification	Defects
Major	Handles or cover fasteners attached to wrong side of case. Cover missing. Control inoperative. Panel warped or bent. Waterproof gaskets damaged or missing.
Minor	Lid fastener missing or inoperative. Panel fastener missing or inoperative.

4.12 Acceptance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Material. - Material shall be as specified herein.

5.2 General requirements. - The Test Set TS-909 /PPM shall be preserved, packaged and packed in accordance with the provisions of this document as referenced in the bid request or contract.

5.2.1 Dimensional data. - Dimensions of packaging materials in the Bill of Material Table are based on the dimensions of the equipment cited in the Table. When the dimensions of the equipment vary from those cited, the dimensions of the packaging materials shall be adjusted accordingly.

5.3 Preservation and Packaging. -

5.3.1 Level A. - Each Test Set TS-909/PPM shall be packaged in accordance with items 1 through 7 of Figure 1 and Table IV.

5.3.2 Level C. - Each Test Set TS-909/PPM shall be packaged as specified in 5.3.1 except item 3 shall be Grade 4 and Class 2, Items 4, 5 and 6 shall be Class 2 and Item 7 shall be Class 1, 175 PSI and the box closure shall be with tape conforming to Specification UU-T-111, Class 2.

5.4 Packing. -

5.4.1 Level A. - Each Test Set TS-909/PPM shall be packed in accordance with Items 8 and 9 of Figure 1 and Table IV. The box closure shall be in accordance with the Appendix of the applicable box specification.

5.4.2 Level B. - Each Test Set TS-909/PPM shall be packed as specified in 5.4.1, except Item 8 shall be Class 1 and 175 PSI. Item 9 shall be omitted.

5.4.3 Level C. - Each Test Set TS-909/PPM shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.5 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

5.6 Package testing. - Interior packages will be tested to assure compliance with the requirements specified herein. The rough handling test or cyclic exposure test will be performed only when invoked in the bid request or contract.

5.7 Workmanship. - The quality of workmanship shall be such as to permit acceptance of the completed preservation, packaging, packing and marking requirements in accordance with the testing specified in 4.12.

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TEST SET TS-909/PPM

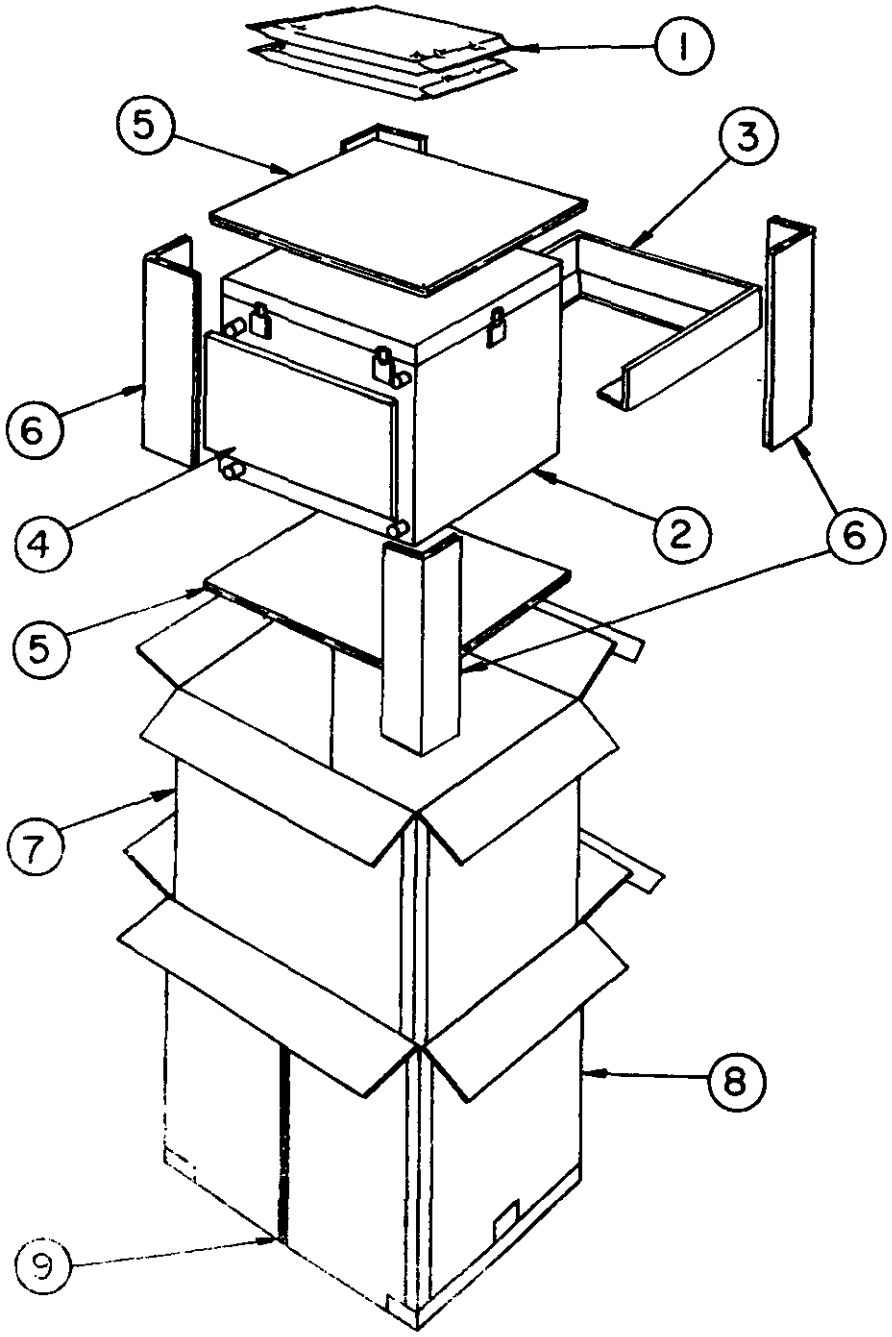


FIGURE 1

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BILL OF MATERIAL
TABLE IX

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Manual	2	IC-6										
	Bag	2		11-1/2	9		MIL-B-117	II	d				
2	Test Set TS-909/PPM	1	IC-5	12-1/2	10-1/2	13-5/8							
	Repair Parts and associated Cables	1 Set		Place in space provided. Secure cables with twine.									
	Twine	3		12	16 ply		T-T-87	I					
3	Sleeve	1		30-1/8	3-1/8		PPP-F-320	CF		SW	W5		
				Score to fit around top and sides of base.									
4	Pad, Back	1		11-1/2	8-3/8	9/16	PPP-F-320	CF		SW	W5		
				Place across back between feet.									
5	Pad, Top and Bottom	2		11-7/8	10	9/16	PPP-F-320	CF		SW	W5		
6	Pad, Corner	4		14-5/8	3	3	PPP-F-320	CF		SW	W5		
				Material 9/16" thick - Score for pads may be made by 90° cut with flutes 1/2 inch deep.									
7	Box, Fiber	1		13-1/8	11-1/4	14-7/8	PPP-B-636	I	2		5	C	RSC
	Tape, Pressure-Sensitive	2		19-1/2	3		PPP-T-76						
		4		17-1/2	3		PPP-T-76						
		1		15	3		PPP-T-76						
8	Box, Fiber	1		13-1/2	11-5/8	15-3/4	PPP-B-636	I	2		3	C	RSC
9	Tape, Filament	1		64	1/2		PPP-T-97	IV					

Lev. A - PACKED WEIGHT 37 Lbs.
Lev. A - PACKED VOLUME 1.53 Cu. Ft.

T - Type V - Variety F - Flute
C - Class G - Grade S - Style

SA-SC Form 2176 Rev
1 Oct 1961

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction inspection

(1) See 3.2.
(2) *Preproduction pack(s) as follows:*

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) *Technical literature required (see 3.9), and quantity of tools and* runing spare parts required.

(h) If rough handling and functional tests are required.

6.2 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TS-909W/PPM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.3 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance Provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product,

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6.4 Deviations.- Request for deviations shall be supported by test data proving the ability of the equipment packaged and packed in accordance with the proposed design change, to withstand the rough handling test requirements of MIL-P-116. Approval will take into consideration the effect on standardization of techniques, availability of packaging and packing materials and economy.

6.5 Options.- Corrugated fiberboard manufactured with A, B or C fluting may be used at the option of the contractor. When the preferred flute is not the same as that cited in the Bill of Material Table, the dimensions of the affected packaging and packing materials will be adjusted accordingly.

Custodian:
Army-EL

Preparing Activity:
Army-EL

Project No. 6625-A 195

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U. S. ARMY ELECTRONICS Command
FORT MONMOUTH, NEW JERSEY 07703

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding Officer
U. S. Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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